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Novel alkali free bioactive fluorapatite glass ceramics

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ABSTRACT

Alkali free bioactive glasses with the presence of Ca/SrF_2 were synthesized and then characterised by Differential Scanning Calorimetry, X-ray Diffraction, Fourier Transform Infrared Spectroscopy and Magic Angle Spinning–Nuclear Magnetic Resonance Spectroscopy. The crystallisation of the glasses was explored by heat treatment. An increased crystallisation tendency of the as quenched glasses was evident with an increase in the Ca/SrF_2 content, where $Ca_{10}(PO_4)_6F_2$ (fluorapatite), $Ca_4Si_2O_7F_2$ (cuspidine) and CaF_2 crystallise in sequence for the Ca-containing system, and $Sr_{10}(PO_4)_6F_2$, $SrSiO_3$ and SrF_2 for the Sr-containing system. Upon heat treatment, $Ca_{10}(PO_4)_6F_2$ or $Sr_{10}(PO_4)_6F_2$, which is mainly nucleated from bulk is the primary crystalline phase. Although partially crystallised these sodium free fluorapatite glass ceramics are proposed to have enhanced bioactivity comparable to bioactive glasses.

1. Introduction

Bioactive glasses and glass ceramics with the ability to form apatite-like phases on heat treatment or on being exposed in physiological solution have been used extensively in medicine and dentistry [1,2]. Apatite glass ceramics are attractive as bone fillers and implant materials, as a result of their favourable biocompatibility and osteoconductivity arising from the presence of apatite [3,4]. Kokubo et al. [5] developed an apatite wollastonite (AW) glass ceramics system, which has high mechanical strength and shows good chemical bonding with living tissue. However, as this system surface nucleates, it has to be sintered and then machined to shape, which is unattractive commercially. In addition, the extremely low CaF_2 content (<1 mol%) used in AW formulations results in a restricted amount of fluoride substitution in apatite and mixed oxy-fluorapatite crystals formation [3].

The castable apatite mullite (AM) glass ceramics based on the $CaO–Al_2O_3–SiO_2–P_2O_5–CaF_2$ system synthesized by Rafferty et al. [6] predominantly bulk nucleate and crystallise to fluorapatite (FAP) and mullite crystals. These bioactive glass ceramics can have high fracture toughness and strength and as a consequence of exhibiting bulk nucleation can be readily cast to shape. Nevertheless, the possibility of toxicity caused by the release of aluminium in vivo is perceived as being problematic for AM glass ceramics.

Brauer and co-workers [7,8] successfully developed two glass ceramic series with different amounts of P_2O_5 in the system of $Na_2O–CaO–SiO_2–P_2O_5–CaF_2$. Mixed sodium calcium fluoride phosphate, FAP, and combeite ($Na_2Ca_2Si_3O_9$) were found as the main crystalline phases in the glasses with high P_2O_5 content (>4.7 mol%) [7], whilst no FAP crystallised from low P_2O_5 (<1.1 mol%) containing glasses [8]. Brauer et al. also found that a sodium free high P_2O_5 containing bioactive glass composition crystallised to FAP. Thus, it is clear that the crystallisation of apatite from glass is strongly influenced by the phosphate content. The presence of sodium is often considered to be essential for bioactive glasses to achieve bioactivity. However, recently, apatite formation in simulated body fluid was also found for glass and glass ceramics that contain no sodium [9–11]. Furthermore, sodium free formulations of bioactive sol–gel glasses have been known for a long time [12] in addition to recently developed hybrid materials [13]. This suggests that sodium is not an essential component for bioactive glasses and glass ceramics [14–16]. In fact high sodium content can cause undesirable rapid rise in pH and a cytotoxic effect.

Fluorapatite ($Ca_{10}(PO_4)_6F_2$) is analogous to hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$; HAP), which is close to the main mineral component of bones and teeth [17,18]. However, FAP is much more stable in acidic environment (pH < 4) compared to HAP, this is extremely favourable in dentistry.

The Sr^{2+} cation is chemically similar to the Ca^{2+} cation but slightly larger in size (effective ionic radius: 1.16 vs 0.94 Å), and therefore it can potentially replace Ca^{2+} in the apatite crystal lattice to form strontium fluoride apatite ($Sr_{10}(PO_4)_6F_2$) [19]. Strontium has the capacity to facilitate enamel remineralisation [20], and the incorporation of strontium into bioactive glasses was reported to promote osteoblast proliferation and reduce osteoclast resorbtion [21,22]. Hence, the strontium containing bioactive glasses are attractive for dental remineralisation and bone regeneration.

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To minimize the disadvantages caused by the presence of sodium and also leverage the benefits by introducing fluoride into glasses, the aim of this paper is to investigate the influence of CaF_2 and SrF_2 content on the crystallisation behaviour of the novel sodium free, high phosphate content bioactive glasses.

2. Materials and methods

2.1. Glass synthesis

Novel sodium free fluoride containing bioactive glasses were designed by introducing CaF_2/SrF_2 into the SiO_2–P_2O_5–CaO ternary glass system (Table 1). Glasses were synthesized by a melt–quench method with analytical grade silica (Prince Minerals Ltd., Stoke-on-Trent, UK), calcium carbonate, phosphorus pentoxide and calcium fluoride or strontium fluoride (all Sigma-Aldrich, Gillingham, UK). A 200 g batch was melted in a 300 ml platinum/rhodium crucible for 1 h in an electrical furnace (EHF 17/3, Lenton, UK). The melting temperature was 1550 °C for glasses with 0 and 3 mol% CaF_2 and 1500 °C for the rest. The melted glasses were quenched into cold deionised water to suppress crystallisation. The as quenched granular glass frit was dried overnight and ground into powder by a vibratory mill (Gyro mill, Glen Creston, UK) for 14 min. The resultant powder was sieved through a 45 μm test sieve (Endecotts Ltd, UK) to obtain fine powder.

2.2. Glass characterisation

2.2.1. X-ray diffraction (XRD)

The amorphous state of the glasses synthesised in the present study was characterised by an X'Pert Pro X-ray diffractometer (PANalytical, The Netherlands) with a copper (Ni-filtered Cu-Kα) X-ray source. The XRD patterns of the glasses were recorded between 5 and 70° 2θ with a step size of 0.0334° and a step time of 200.03 s. Phase identification was carried out by using X'Pert HighScore Plus (v2.0, PANalytical, The Netherlands) in conjunction with the ICDD PDF-4 database.

2.2.2. Fourier Transform Infrared Spectroscopy (FTIR)

Attenuated Total Reflectance–Fourier Transform Infrared Spectroscopy (Spectrum GX, Perkin-Elmer, USA) was used to analyse the chemical bonding in the glass powder and heat treated samples. Ten scans were performed for each sample, and the data were collected for wavenumbers from 500 to 1600 cm^{-1}.

2.2.3. Magic Angle Spinning–Nuclear Magnetic Resonance (MAS–NMR)

^{31}P and ^{19}F MAS–NMR were employed to investigate glass structure and crystalline phases by using a 600 MHz (14.1T) Bruker solid state NMR Spectrometer. Experiments were carried out at spinning rates of 18 or 21 kHz and resonance frequencies of 242.9 and 564.7 MHz for ^{31}P and ^{19}F MAS–NMR, respectively. Recycle delay time was 60 s for ^{31}P and 30 s for ^{19}F. Typically 16 or 32 scans were used for ^{31}P, whilst 32 or 64 scans were adopted for ^{19}F MAS–NMR. Chemical shift for ^{31}P and ^{19}F was referenced by using 85% H_3PO_4 and 1 M NaF solution; the latter signal was adjusted to −120 ppm relative to $CFCl_3$.

2.2.4. Differential Scanning Calorimetry (DSC)

Differential Scanning Calorimetry (DSC) was used to characterise the thermal properties of the glasses. Glass frit or fine powder (50 ± 1 mg) were heated under Nitrogen (60 ml/min^{-1}) at a rate of 20 °C/min from room temperature up to 1100 °C against alumina using a differential scanning calorimeter (DSC 1500 Stanton Redcroft, Rheometric Scientific, UK). Glass transition temperature (T_g) and crystallisation peak temperatures (T_p) of each glasses were then obtained from the recorded DSC trace. The accuracy of all characteristic temperatures obtained and further presented in figures and tables was ± 5 °C.

2.3. Crystallisation behaviour

1 g of fine powder (<45 μm) was compacted into a disc, placed on a platinum foil and heated up at a rate of 20 °C/min from 400 °C to the heat treatment temperature (T_{HT}) in a porcelain furnace (CeramPress, NEY Dental International, USA) without holding. According to the width of the crystallisation peak from the DSC traces, T_{HT} was T_p or approximately 10 °C above T_p. After heat treatment, samples were immediately withdrawn from the furnace and air quenched.

The resulting sintered glass ceramics were then ground into powder and characterised by using XRD, FTIR and NMR. In order to investigate the morphology of the crystalline phases, the heat treated glass ceramics were embedded in resin, polished with a series of silicon carbide grinding paper and finished with 0.3 μm diamond paste (Met Prep Ltd., UK). The polished specimens were gold coated and viewed under a scanning electron microscope (FEI Inspect F, Oxford Instruments, UK) using back-scatter and secondary electron imaging modes.

3. Results

3.1. Glass formation

The XRD patterns of the as quenched sodium free Ca- and Sr-containing glasses are given in Fig. 1a and b. Amorphous halos at 30° 2θ were found in glasses with low CaF_2 content (less than 9.3 mol%), whereas clear diffraction peaks were noticed for higher CaF_2 containing glasses. With an increase in the CaF_2 content (up to 9.3 mol%), additional small features at 25.86°, 32.23° and 33.07° 2θ were evident. These peaks were matched to the major apatite (FAP: 00-034-0011) diffraction lines. However, a minor peak at 31.89° 2θ was seen for glasses with 4.5 and 6.0 mol% CaF_2. A further increase in the CaF_2 content up to 17.8 mol% resulted in the crystallisation of cuspidine ($Ca_4Si_2O_7F_2$, 00-011-0075) with diffraction lines at 26.5° and 29.1°. At the same time, additional lines at 28.28° and 47° 2θ corresponding to fluorite (CaF_2, 00-004-0864) were also found.

The SrF_2-based series demonstrated similar patterns (Fig. 1b). Amorphous halos were exhibited at 29° 2θ in the glass with SrF_2 contents less than 4.5 mol%, and sharp peaks at 26.5°, 27.85°, 30.5° and 31.7° 2θ were observed for glasses with SrF_2 content greater than 4.5 mol%, which were identified with the main diffraction lines of $Sr_{10}(PO_4)_6F_2$ (00-050-1744). With increasing SrF_2 content up to 13.6 mol%, peaks corresponding to $SrSiO_3$ (00-034-0099) appeared at 24.95° and 43.9° 2θ. Furthermore, additional peaks at 43.19° and 52.23° in the XRD patterns of 17.8 and 25.5 mol% SrF_2 glasses were matched to the characteristic diffraction peaks of SrF_2 (00-001-0644). The crystalline phases identified in both glass series are summarised in Table 2.

FTIR spectra of the as quenched CaF_2 containing glasses (Fig. 2a) show the typical non-bridging oxygen band and Si–O–Si stretch band at 920 and 1030 cm^{-1} for the whole series. In addition, a small band around 600 cm^{-1} was noticed in the glasses with CaF_2 content lower

Table 1
Glass compositions in mol%. GF is used for CaO/CaF_2 series and SF for SrO/SrF_2 series.

Glass code	SiO_2	CaO/SrO	P_2O_5	CaF_2/SrF_2
GF/SF 0.0	38.1	55.5	6.3	0.0
GF/SF 3.0	37.0	53.9	6.1	3.0
GF/SF 4.5	36.4	53.0	6.0	4.5
GF/SF 6.0	35.9	52.2	5.9	6.0
GF/SF 9.3	34.6	50.4	5.7	9.3
GF/SF 13.6	32.9	48.0	5.5	13.6
GF/SF 17.8	31.4	45.7	5.2	17.8
GF/SF 25.5	28.4	41.4	4.7	25.5

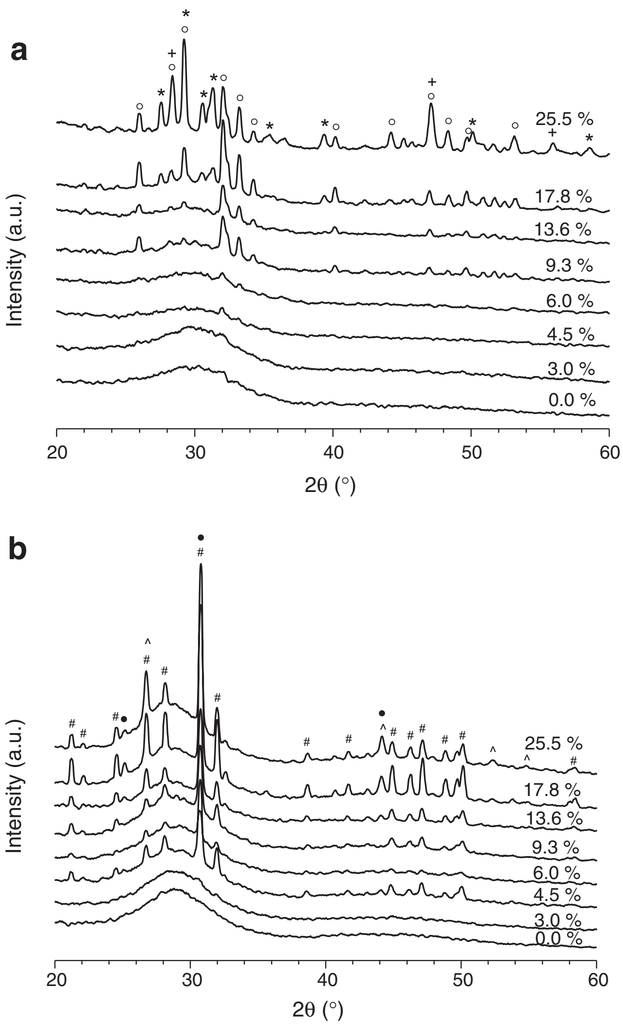

Fig. 1. XRD patterns of the as quenched glasses. (a) CaF_2-based glasses (°: $Ca_{10}(PO_4)_6F_2$; *: $Ca_4Si_2O_7F_2$; +: CaF_2); (b) SrF_2 glasses (#: $Sr_{10}(PO_4)_6F_2$; ^: SrF_2; ●: $SrSiO_3$). The numbers are nominal molar percentage of the CaF_2/SrF_2 in the compositions.

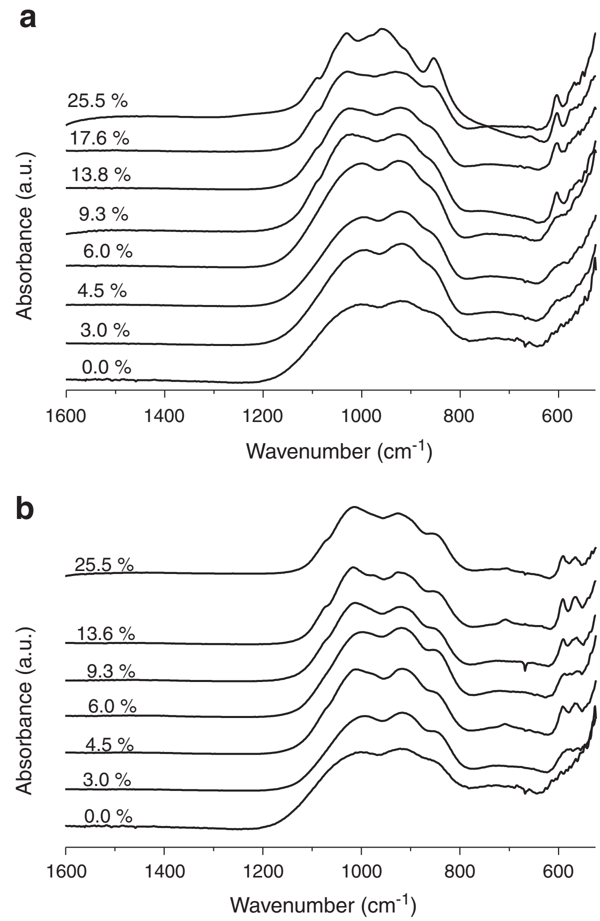

Fig. 2. FTIR spectra of the as quenched glasses. (a) CaF_2 glasses; (b) SrF_2 glasses. The numbers are nominal molar percentage of the CaF_2/SrF_2 in the compositions.

than 9.3 mol%, whilst the characteristic apatite split bands at 560 and 613 cm^{-1} were visible with an increase in CaF_2 content [23]. The spectrum of the Ca-containing glass with 25.5 mol% of CaF_2 was characterised with distinctly sharp band features in the region 800–1050 cm^{-1} indicating presence of silicate crystalline phase [24].

SrF_2 containing glasses demonstrated a similar trend as CaF_2 series (Fig. 2b). However, the typical apatite split bands at 595 cm^{-1} and 555 cm^{-1} were first recognised in the strontium series with 4.5 mol% SrF_2 content. At the same time the spectra of the glasses with high fluoride content clearly showed presence of a silicate crystalline phase.

^{31}P and ^{19}F MAS–NMR were used to characterise the atomic environment of phosphate and fluoride in the glasses. Fig. 3a shows the ^{31}P MAS–NMR spectra of CaF_2 containing bioactive glasses and glass ceramics. Broad peaks with a centre at 3 ppm, corresponding to amorphous calcium orthophosphate, are found in the spectra of low CaF_2 containing glasses. In addition, with an increasing amount of CaF_2 the linewidth of the peak reduced, but without changing the peak position. This suggests that glasses tend to partially crystallise to calcium orthophosphate of which, apatite is an example, which leads to a reduction in the linewidth of the ^{31}P NMR signal.

Three types of fluoride environment were presented by ^{19}F MAS–NMR spectra in Fig. 3b. The glasses with low CaF_2 content have broad peaks with a centre at about −96 ppm, whilst a tiny but sharp peak at −103 ppm, corresponding to fluorapatite is found on the shoulder of a broad glass signal in the glasses with CaF_2 content over 9.3 mol%. Furthermore, an asymmetric and sharp peak at −108 ppm of the chemical shift is obvious in the spectrum of the 25.5 mol% CaF_2 containing glass, indicating the crystallisation of CaF_2.

Table 2
The summary of crystalline phases identified from XRD patterns.

Glasses	Crystallisation phases	Glasses	Crystallisation phases
GF 0.0		SF 0.0	Amorphous
GF 3.0	Amorphous	SF 3.0	
GF 4.5		SF 4.5	
GF 6.0		SF 6.0	$Sr_{10}(PO_4)_6F_2$
GF 9.3	$Ca_{10}(PO_4)_6F_2$	SF 9.3	
GF 13.6		SF 13.6	$Sr_{10}(PO_4)_6F_2$, $SrSiO_3$
GF 17.8	$Ca_{10}(PO_4)_6F_2$, $Ca_4Si_2O_7F_2$, CaF_2	SF 17.8	$Sr_{10}(PO_4)_6F_2$, $SrSiO_3$, SrF_2
GF 25.5		SF 25.5	

3.2. Thermal properties

The glass transition temperature and crystallisation peak temperatures for both glass powder and frit are summarised and listed in Tables 3 and S1 (Supplemental material). Some of the compositions had more than one exotherm. The corresponding crystallisation peak temperatures, of the first (T_{p1}), second (T_{p2}), third (T_{p3}) and fourth (T_{p4}) exotherms were defined from the DSC traces. Particle size had no significant effect on T_g and some of the crystallisation temperatures, which result in close values for T_g, T_{p1} and some of the T_{p2} and T_{p3} values. However, some crystallisation peak temperatures of coarse

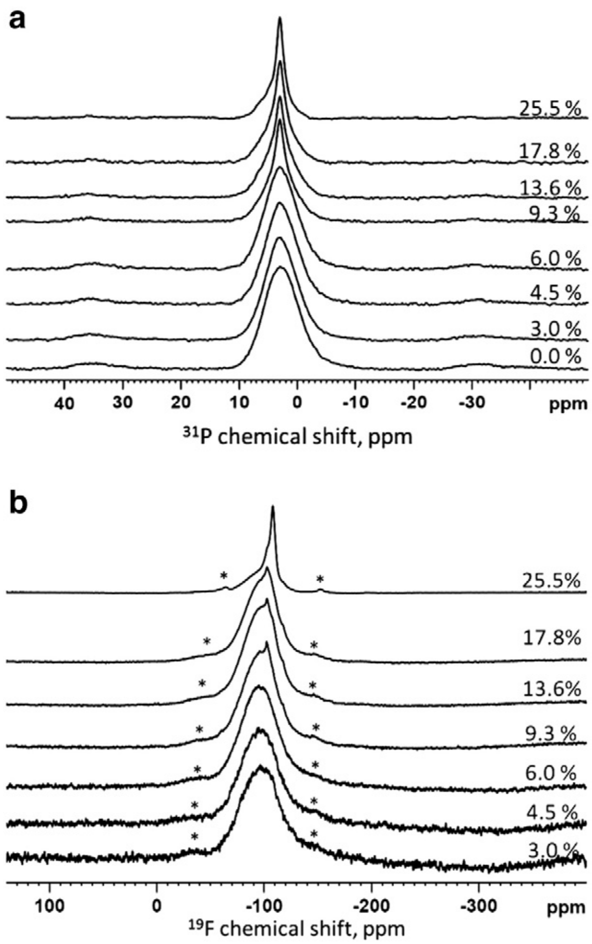

Fig. 3. (a) ^{31}P MAS–NMR and (b) ^{19}F MAS–NMR spectra of CaF_2-based glasses. The asterisk shows the spinning side bands. The numbers are nominal molar percentage of the CaF_2 in the compositions.

particles (frit) are higher than the equivalent fine powder, which were highlighted in Table 3.

As shown in Fig. 4, the glass transition temperature and first crystallisation peak temperature showed a similar trend, both reducing with increasing CaF_2 or SrF_2 content. Comparing T_g and T_{p1} of the two glass series, the T_{p1} values for the SrF_2 glasses are much lower (30–70 °C) than the equivalent ones for the CaF_2 series, though the T_g values were about the same.

3.3. Crystallisation behaviour

XRD results of heat treated glasses GF 3.0 and GF 25.5 are presented in Fig. 5. For GF 3.0 after heat treatment at T_{p1} (867 °C), sharp diffraction

Table 3
Characteristic temperature of glass powder (<45 μm) for fluoride containing glasses. The glasses mainly nucleated form surface are highlighted in bold.

XF_2 (mol%)	Glass powder (CaF_2 series)				Glass powder (SrF_2 series)				
	T_g (°C)	T_{p1}	T_{p2}	T_{p3}	T_g (°C)	T_{p1}	T_{p2}	T_{p3}	T_{p4}
GF 0.0	790	**925**	–	–	783	918	1042	–	–
GF 3.0	735	867	888	979	735	821	**862**	**955**	848
GF 4.5	722	860	–	–	706	801	**827**	**918**	–
GF 6.0	693	817	**844**	942	680	778	**795**	**859**	–
GF 9.3	655	780	**806**	**878**	657	744	–	862	–
GF 13.6	618	732	758	977	618	705	**716**	853	–
GF 17.8	601	720	–	–	–	–	–	–	–
GF 25.5	558	713	–	–	562	638	699	713	847

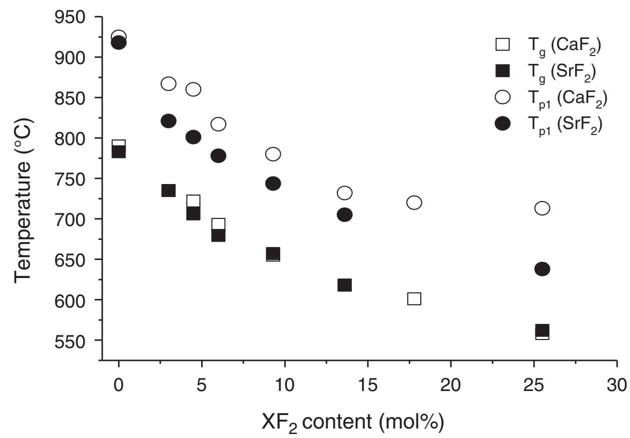

Fig. 4. Glass transition temperature (T_g) and first crystallisation peak temperature (T_{p1}) of fine powder against fluoride content. The accuracy of the defined temperature values is ±5 °C.

lines corresponding to apatite develop, and there is a trace of wollastonite present as well. Interestingly no new crystalline phases were found for the heat treatments at the higher temperatures. The partially crystallised glass GF 25.5, that already contains apatite, cuspidine and fluorite shows increase in diffraction line intensity of in particular cuspidine and fluorite indicating further crystallisation on heat treatment at 713 °C. The crystal phases formed on heat treatment of the glasses are presented in Table 4. Fig. S1 (Supplemental material) shows the XRD pattern of GF 0.0 heat treated at 1030 °C. Compared to

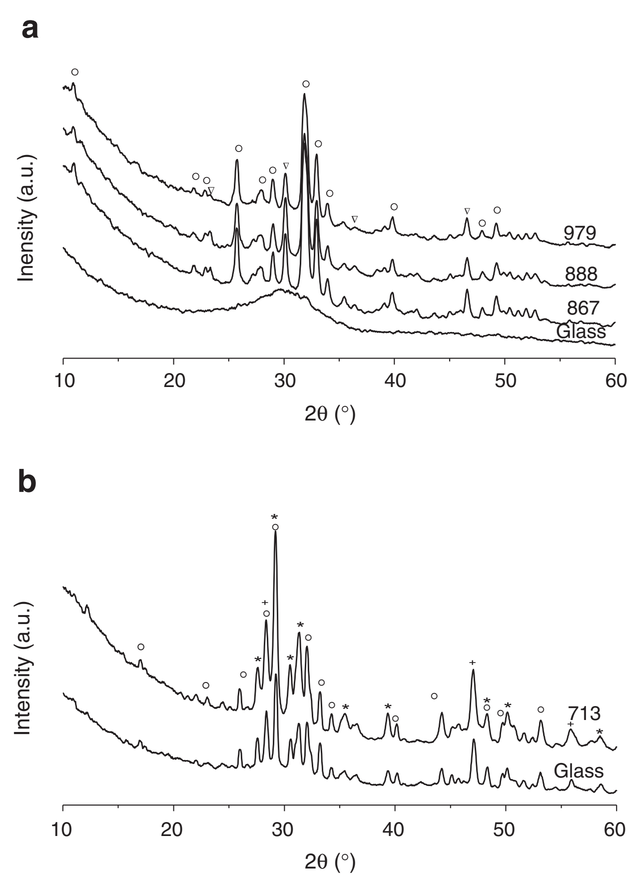

Fig. 5. XRD patterns of (a) GF 3.0 and (b) GF 25.5 as-quenched and heat treated at different temperatures. (∇: $CaSiO_3$, o: $Ca_{10}(PO_4)_6F_2$; *: $Ca_4Si_2O_7F_2$; +: CaF_2). The numbers are heat treatment temperature.

Table 4
Summary of phases crystallising on heat treatment. β-$CaSiO_3$: pseudowollastonite, $Ca_{10}(PO_4)_6(OH)_2$: hydroxyapatite, $Ca_{10}(PO_4)_6F_2$: fluorapatite, $CaSiO_3$: wollastonite-2M, $CaSiO_3$ (hexagonal): wollastonite-1A, $Ca_4Si_2O_7F_2$: cuspidine, CaF_2: fluorite, $SrSiO_3$: strontium silicate, $Sr_{10}(PO_4)_6F_2$: strontium phosphate fluoride and SrF_2: strontium fluoride.

Glasses	T_{HT} (°C)	Crystalline Phases
GF 0.0	1030	β-$CaSiO_3$ and $Ca_{10}(PO_4)_6(OH)_2$
GF 3.0	867	$Ca_{10}(PO_4)_6F_2$ and $CaSiO_3$
	888	
	979	
GF 4.5	860	
GF 6.0	817	
	844	
	942	$Ca_{10}(PO_4)_6F_2$ and $CaSiO_3$ (hexagonal)
GF 9.3	780	$Ca_{10}(PO_4)_6F_2$
	806	$Ca_{10}(PO_4)_6F_2$ and $Ca_4Si_2O_7F_2$
	878	
GF 13.6	732	
	758	
	977	
GF 17.8	720	$Ca_{10}(PO_4)_6F_2$, $Ca_4Si_2O_7F_2$ and CaF_2
GF 25.5	713	
SF 3.0	1040	$SrSiO_3$ and $Sr_{10}(PO_4)_6F_2$
SF 4.5	860	
	1000	
SF 6.0	830	$SrSiO_3$, $Sr_{10}(PO_4)_6F_2$ and SrF_2
	960	

the amorphous halo of the initial glass, distinct diffraction lines, which match with hydroxyapatite and pseudowollastonite are evidenced.

4. Discussion

4.1. Glass formation

XRD patterns, FTIR and NMR spectra all demonstrated that the low fluoride content glasses were amorphous, but the glasses with higher CaF_2 (≥9.3 mol%) and SrF_2 (≥4.5 mol%) content were partially crystallised to FAP ($Ca_{10}(PO_4)_6F_2$, and $Sr_{10}(PO_4)_6F_2$). The GF 13.6 glass crystallised to FAP and cuspidine. Crystalline CaF_2 and SrF_2 were found in addition on increasing the fluoride content to over 17.8 mol% CaF_2 and 13.6 mol% SrF_2, respectively. Interestingly, based on XRD results, the highest CaF_2 content (25.5 mol%) glass demonstrated a pronounced increase in the intensity of the diffraction lines corresponding to cuspidine, and this increase was at the expense of the intensity of the lines corresponding to FAP compared with the GF 17.8.

The SrF_2 containing glasses had a higher tendency to crystallise than CaF_2 ones. This can be attributed to the more expanded glass structure causing by the larger size of Sr^{2+} [22,25].

^{31}P MAS–NMR spectra suggest that in the sodium free fluoride containing bioactive glasses and glass ceramics, phosphorus is present as a calcium orthophosphate. ^{19}F MAS–NMR confirmed the presence of amorphous F–Ca(n) species and also identified spontaneous FAP (F–Ca(3)) crystallisation in the glasses with over 9.3 mol% CaF_2 content on quenching, whilst CaF_2 with a F–Ca(4) environment crystallised spontaneously from the glass with 25.5 mol% CaF_2.

Cuspidine has been found to crystallise from sodium and fluoride-containing bioactive glasses [8], glasses in the $CaO–CaF_2–SiO_2$ system [26,27] and mould-flux glasses [28]. Our study suggests that for glasses in the $CaO–CaF_2–P_2O_5–SiO_2$ system with a constant $SiO_2:CaO:P_2O_5$ molar ratio of 6:8.8:1, an increase in the CaF_2 content leads to an increased crystallisation tendency, where FAP, cuspidine and CaF_2 crystallise sequentially. At moderate fluoride content (at 9.3 mol%) only FAP forms in the as quenched glass, with increasing in fluoride content, two fluoride containing phases are present in the as quenched glass, i.e. FAP and cuspidine. The glasses with the highest fluoride content have all three fluoride containing phases.

The minute amount of apatite seen in the ground glass powders with the lowest fluoride content is due to the reaction with atmospheric moisture.

4.2. Thermal properties

Owing to fluoride complexing calcium, glass transition and crystallisation temperatures decreased markedly with increasing CaF_2/SrF_2 content (0 to 25.5 mol%) [8]. In the fluoride free glass (0 mol%), divalent calcium ions bind together silicate anions by electrostatic forces and the calcium ions effectively act as ionic bridges between two non-bridging oxygens (NBOs). When CaF_2/SrF_2 is added, hypothetical 'CaF^+'/'SrF^+' species are formed, and the electrostatic forces between NBOs are reduced considerably, resulting in a decrease in T_g [8,29]. The effect of CaF_2/SrF_2 on T_g was less pronounced at high fluoride content (Fig. 4). This could be attributed to the crystallisation of fluoride containing phases (such as FAP and CaF_2/SrF_2) in the as quenched glasses. However, as Sr^{2+} is larger in size compared to Ca^{2+}, it is thought to expand the glass network more [22,25], resulting in a less stable glass which crystallises more readily and at a lower temperature.

The glass transition temperature of the fluoride free glass (0 mol%, 790 °C) was considerably higher than that of the equivalent soda containing glass (ICSW9, 491 °C) studied by O'Donnell et al. [30]. This can be explained by differences in cross-linking. Owing to Ca^{2+} being replaced for two Na^+ (i.e. a reduced charge-to-size ratio of the modifier cation), ionic cross-linking between NBOs of neighbouring silicate chains is reduced, and as a consequence T_g increases. T_g is determined by the strength of the bonding in the glass and is also known to correlate with degradability and hardness of glasses [31]. In the absence of sodium, the addition of fluoride is particularly beneficial since fluoride decreases T_g and melting temperature in a similar fashion, facilitates melting and is expected to reduce glass hardness.

In Tables 3 and S1 (Supplemental material), the similar first crystallisation peak temperatures (T_{p1}) of powder and frit indicate that in the CaF_2 and SrF_2 containing glasses the first crystalline phases, which are believed to be $Ca_{10}(PO_4)_6F_2$ or $Sr_{10}(PO_4)_6F_2$, respectively, are mainly nucleated from bulk. Whilst T_{p1} of the frit is higher than fine powder for GF 0.0 (fluoride free), this is ascribed to wollastonite, which is a phase known to strongly surface nucleate. The second or third crystallisation phases of some compositions are also strongly nucleating from the surface. They are highlighted in Table 3. Compared with frit, fine powder has larger surface area, which provides more nucleation sites for crystallisation. Therefore fine powder is prone to crystallise at a lower temperature [32]. In contrast, when the crystallisation peak temperature is independent of particle size, then this may indicate bulk nucleation.

Crystallisation of a bioactive glass has been shown to reduce its bioactive properties [33] as the crystalline phases are generally more stable than the equivalent glass phase and generally crystallisation results in a residual glass phase with a higher network connectivity (NC). However, in the present glasses, crystals of FAP ($Ca_{10}(PO_4)_6F_2$ and $Sr_{10}(PO_4)_6F_2$) are able to serve as nuclei for the further growth of the apatite phase, which may therefore increase its bioactivity. In addition, the apatite formation will not result in a less reactive residual silicate glass since the NC of the glass will remain unchanged. The bioactivity of these glasses and glass-ceramics has been presented and discussed in a separate report [34].

4.3. Glass heat treatment

Two or three crystal phases were typically found upon heat treatment. The first was apatite, which is thought to be HAP, presenting as a hexagonal-like morphology in the fluoride free composition. Pseudowollastonite (β-$CaSiO_3$) also formed at the heat treatment temperature of 1030 °C (Fig. S1, Supplemental material). It is surprising that an apatite was formed in the absence of fluoride since in AM glass ceramics, apatite only forms when fluoride is present [6]. It may be that the apatite formed is an oxyapatite which transforms to HAP by reaction with atmospheric water. The crystallisation peak observed for this composition was broad and is probably a result of the overlapping of apatite and pseudowollastonite crystallisation events.

GF 3.0 crystallised to FAP with an elongated and interlocking crystal structure (dendritic) (Fig. S2, Supplemental material) at 888 °C with coexisting FAP and wollastonite phases. All the heat treated samples of 4.5 and 6.0 mol% CaF_2 containing glasses crystallised to FAP and wollastonite. The heat treatment temperatures are too high to see the crystallisation of FAP without wollastonite. The wollastonite phase was expected on the grounds that the residual glass phase following crystallisation of FAP would have a Q structure close to Q^2 and also close to the stoichiometry of wollastonite.

FAP, cuspidine and CaF_2 are found as crystalline phases in the glasses with over 6 mol% CaF_2 following heat treatment. It is worth noting that in these series of glasses the Q structure did not change on crystallisation and remained at 2.08 (NC) irrespective of how much apatite crystallises. Cuspidine and CaF_2 are phosphate free crystalline phases and therefore should not restrict bioactivity. This is an attractive feature of these glasses and glass ceramics, since the essentially Q^2 speciation of the amorphous structure part will confer bioactivity. In addition, the presence of FAP is also extremely desirable and FAP glass ceramics have demonstrated excellent osseointegration [35]. The FAP may also provide seed crystals for the formation of more apatite as the Q^2 residual glass phase dissolves in body fluids. These glass ceramics combine the beneficial effects of a bioactive glass phase and the FAP crystals.

5. Conclusion

This study revealed that novel sodium free low fluoride content ($CaF_2 < 9.3$ mol% and $SrF_2 < 4.5$ mol%) bioactive glasses can be obtained in an amorphous state. Bioactive glasses with high fluoride content ($CaF_2 \geq 9.3$ mol% and $SrF_2 \geq 4.5$ mol%) crystallised spontaneously to fluorapatite during quenching, whilst CaF_2 and SrF_2 were found in the glasses with 17.8 and 25.5 mol% fluoride. Fluorapatite crystallised from all glasses upon heat treatment. An addition of fluoride into bioactive glasses leads to a dramatic reduction in T_g and T_{p1}. In order to obtain the glass ceramics for dental applications which show FAP as the only crystal phase, large crystallisation windows and optimised fluoride content (about 6.0 mol% CaF_2 and 4.5 mol% SrF_2) are required.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.jnoncrysol.2014.05.025.

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